Before Any Drug. Feeds, Thickeners, the Milk Trial and Safe Sleep in the Regurgitating Infant.

Picture a two-month-old boy brought to clinic by parents who are washing three loads of laundry a day. He brings up milk after most feeds, sometimes a lot of it. He is gaining weight well, he has no warning signs, and in the previous lesson we learned that this is a happy spitter whose regurgitation will peak at around four months and disappear in nine out of ten infants by the first birthday. The parents accept that he is healthy. But they still want help, and they have heard about a reflux medicine. They ask what they can do instead.

That is the clinical question of this chapter. In an infant with troublesome regurgitation, which non-drug measures actually work, how well, and at what risk? The main idea is this. Non-drug measures treat the spilling and the family's distress, not a disease. Thickening the feed reduces the milk that comes out of the mouth, but it barely changes how often reflux happens inside the oesophagus. Nothing on the list has been shown to reduce crying. A trial of removing cow's milk protein can uncover a real allergy, but only if it is run as a proper test with a re-challenge. And the one measure that families most often try, raising the head of the bed or using an inclined sleeper, does not work and can kill. Each well-run week of non-drug care is also a week without a proton pump inhibitor, a drug that in infants matched placebo for symptoms and caused more infections.

A word on sources. The main document is the 2018 joint guideline on gastro-oesophageal reflux in infants and children from the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, called NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, called ESPGHAN, written by Rosen and colleagues. Alongside it sit the British guideline from the National Institute for Health and Care Excellence, called NICE, numbered NG1 and updated in 2019, with its quality standard from 2016. On sleep, the source is the 2022 policy statement of the American Academy of Pediatrics, called the AAP, written by Moon and colleagues. On cow's milk allergy, it is the 2024 ESPGHAN position paper written by Vandenplas and colleagues. The evidence on thickening comes from a Cochrane review by Kwok and colleagues in 2017, a meta-analysis by Horvath and colleagues in 2008, and a physiological study by Wenzl and colleagues in 2003. On positioning, we draw on a trial by Loots and colleagues in 2014 and an older study by Orenstein and Whitington in 1983.

Part one. Where each measure acts.

To understand why some measures help and others do not, follow a single reflux event step by step.

The first step is distension. After a large feed the upper part of the stomach stretches. That stretch activates nerve endings that send signals up the vagus nerve to the brainstem and back

down again. The result is the second step: a transient lower oesophageal sphincter relaxation, which we abbreviate as TLESR. This is a deliberate opening of the valve between the stomach and the oesophagus, not a weakness of the valve. In infants, between eighty-two and ninety-four percent of reflux episodes happen during one of these relaxations. Here is the first important point: no thickener and no body position stops a TLESR. Only a smaller stretch, meaning a smaller feed, reduces the trigger.

The third step is the rise of the refluxed milk up the oesophagus. Thin milk climbs easily, all the way to the mouth. Thickened milk rises less high, so less of it reaches the mouth and is seen as regurgitation. The fourth step is exit, where gravity matters: how the infant is lying decides how much milk spills out. When an infant lies on the left side, the junction between the stomach and oesophagus sits above the pool of milk, so less reflux occurs.

And there is a fifth element that sits across all the steps. In an infant with cow's milk protein allergy, the allergic inflammation can disturb the motility of the stomach and oesophagus and slow gastric emptying, which amplifies every step. Here, removing the protein treats the cause rather than the consequence.

Remember too, from the first lesson of this block, that most infant reflux is not acidic, because milk buffers the stomach for up to two hours after a feed. So non-drug measures change the volume, the height and the visibility of reflux much more than its frequency. That is why they help the spilling and not the crying.

Part two. Feed volume.

Many infants who spit a lot are simply taking a lot. Yet the evidence here is surprisingly thin. When the 2018 joint guideline searched for trials of reduced feed volume or more frequent feeds, not a single study met its inclusion criteria. The advice to avoid overfeeding is therefore based on physiology and expert opinion, not trials.

NICE turns this into a stepped approach for formula-fed infants with frequent regurgitation and marked distress. First, review the feeding history. Second, reduce feed volumes, but only if they are excessive for the infant's weight. Third, offer a trial of smaller, more frequent feeds, while keeping the total daily amount appropriate. Neither document gives a number for what counts as excessive.

In practice, this means asking precise questions. How many millilitres per feed? How many feeds a day? Who feeds at night? Then divide the daily total by the infant's weight. Ask whether a bottle is offered every time the infant cries, because hunger cues and the cues of colic look alike, and a crying infant who is fed every time can easily be overfed. The growth chart is the referee. A spilling infant who is gaining weight well is not underfed. Reduce only what is genuinely excess, divide the same daily total into more feeds, and check the weight again in one to two weeks. And never cut the daily volume of an infant whose growth is faltering just to reduce spilling. The 2018 guideline also asks every clinician to record

exposure to tobacco and second-hand smoke in the history of an infant with reflux symptoms.

Part three. Thickening.

Thickening is the best-studied non-drug measure, and the most instructive. Consider first the physiological study by Wenzl and colleagues, published in the journal Pediatrics in 2003. Fourteen infants, about six weeks old on average, with recurrent regurgitation and no other symptoms, were studied in a placebo-controlled crossover design. Each infant alternated between formula thickened with carob bean gum and ordinary formula across six feeding periods, while impedance and pH monitoring recorded every reflux event and a video camera recorded every regurgitation. Over three hundred and forty-two hours of recording, regurgitation fell from sixty-eight episodes on standard formula to fifteen on thickened formula, a reduction of about seventy-eight percent, and the severity score fell from one point eight to point six. But reflux events detected by impedance fell only from six hundred and forty-seven to five hundred and thirty-six, a reduction of about seventeen percent, and almost all of that reduction was in non-acid reflux. Acid reflux and the duration of episodes did not change, and the height the reflux reached fell, but not significantly. In other words, thickening converts visible regurgitation into silent reflux. It treats the spill, not the reflux.

The pooled trials say the same thing. The 2017 Cochrane review by Kwok and colleagues included eight randomised trials with six hundred and thirty-seven infants up to six months of age. In six trials with four hundred and forty-two infants, thickened feeds reduced episodes of regurgitation or vomiting by about two per day, with a mean difference of minus one point nine seven and a ninety-five percent confidence interval from minus two point three two to minus one point six one. In two trials with one hundred and eighty-six infants, thickened feeds made an infant two and a half times more likely to be free of regurgitation after one to eight weeks, which corresponds to treating five infants for one to become symptom-free. The earlier meta-analysis by Horvath and colleagues, of fourteen trials and eight hundred and seventy-seven infants, found more infants free of regurgitation and slightly fewer episodes of regurgitation and vomiting, a modest increase in daily weight gain, and no serious harms; but it found no effect on the reflux index, which is the percentage of time with acid in the oesophagus, nor on acid reflux episodes per hour. No thickener was clearly better than another. The 2018 joint guideline rates the quality of this evidence as low to very low, and suggests thickened feeds for visible regurgitation and vomiting. NICE offers thickened formula as the fourth step, after volume review and smaller feeds. What none of these trials shows is any effect on crying, arching or sleep.

Part four. Which thickener.

The choice of agent depends on the age of the infant and on the milk. Rice cereal added to formula is cheap and widely available, but it has three problems. It adds energy, so a thickened bottle is also a denser bottle, which can push weight gain upward. It needs a teat with a bigger hole. And rice contains inorganic arsenic. In 2020 the United States Food and Drug

Administration, called the FDA, set an action level of one hundred parts per billion for inorganic arsenic in infant rice cereal, and it advises that oat, barley and multigrain cereals are reasonable alternatives.

Pre-thickened anti-regurgitation formulas, which use rice or corn starch, are thin in the bottle and thicken in the acid of the stomach. They have the same energy content as standard formula, so they do not add calories, and cereal should not be added on top.

Carob bean gum, also called locust bean gum, is a non-digestible gum that thickens milk in the bottle. It can be used with formula or with expressed breast milk, and the guideline sets its lower age limit at forty-two weeks post-menstrual age. It can loosen the stools.

Xanthan gum thickeners need a special warning. In May 2011 the FDA, with an alert from the Centers for Disease Control and Prevention, warned that a xanthan-based thickener had been linked to seventeen cases of necrotising enterocolitis, called NEC, and five deaths in premature infants, some of them after discharge home. The advice was not to give it to infants born preterm. Guidance on reflux restricts xanthan thickeners to children older than one year. This is a real risk for the graduate of the neonatal unit whose family buys a thickener online.

Thickening breast milk with starch rarely works, because the amylase in breast milk digests the starch within minutes. And infant alginate sachets are not really thickeners at all; the alginate forms a gel raft in contact with stomach acid. We will return to their evidence shortly.

Part five. The milk protein overlap.

Cow's milk protein allergy can look exactly like reflux. In an Italian series published in the Archives of Disease in Childhood in 1996, fourteen of forty-seven infants with primary reflux, about thirty percent, also had cow's milk allergy; their mean age was about eight months. At the same time, allergy is over-diagnosed. In the European birth cohort called EuroPrevall, which followed more than twelve thousand newborns, three hundred and fifty-eight infants were referred with suspected cow's milk allergy, but only fifty-five were confirmed by a double-blind food challenge, about one in seven. The challenge-proven incidence was just over half a percent. The 2024 ESPGHAN position paper is explicit that over-diagnosis is far more common than under-diagnosis, and that a long unnecessary diet carries nutritional and social costs.

So when is a milk trial reasonable? When troublesome regurgitation persists despite good feeding advice, or when regurgitation comes with eczema, blood or mucus in the stool, diarrhoea, a strongly atopic family, or slow weight gain. The 2018 joint guideline suggests a two- to four-week trial of an extensively hydrolysed formula, or of removing cow's milk from the mother's diet if the infant is breastfed, in infants whose symptoms persist after optimal non-drug treatment, and before any acid suppression.

A milk trial is a test, so it needs a start, an end and a re-challenge. Select the infant carefully. Eliminate cow's milk protein completely: for a formula-fed infant, change to an extensively hydrolysed formula; for a breastfed infant, the mother removes all dairy from her diet and takes calcium and vitamin D. At two to four weeks, judge the result with a symptom diary. If there is no clear improvement, this is not allergy, and the diet should stop. If there is improvement, re-challenge with cow's milk, at home if the reaction is not mediated by immunoglobulin E and was never severe, or in hospital if there were immediate allergic features. If symptoms return, the diagnosis is made, and the diet continues with a review of tolerance every six to twelve months. If symptoms do not return, the infant goes back to normal feeds. Extensively hydrolysed formula is the first choice; an amino acid formula is reserved for severe reactions or for failure of the hydrolysate. The 2018 guideline does not recommend soy formula or rice hydrolysate for this trial. And throughout, breastfeeding should continue; the maternal diet is the trial, and switching to formula is not.

Part six. Positioning, and why sleep is different.

Body position does change reflux physiology. In the trial by Loots and colleagues in 2014, fifty-one infants aged up to six months with suspected reflux disease were randomised to a proton pump inhibitor or an antacid, each combined with either left lateral positioning or elevation of the head of the cot, with eight-hour impedance and pH studies before and after two weeks. Lying on the left side reduced total reflux episodes by about twenty-one per eight hours, compared with about ten with head elevation, a difference that just missed significance with a p value of point zero five six. The combination of the drug and left lateral positioning reduced reflux episodes from sixty-nine to forty-six, and the drug reduced acid exposure more than antacid. But on standard analysis, no group improved in crying or irritability; only vomiting fell, in the antacid with left lateral group. An older study from 1983 showed that sitting an infant in an infant seat after feeds caused more reflux than prone, head-elevated positioning, probably because a slumped, semi-upright posture compresses the stomach.

So why do we not recommend positioning? Because sleeping on the side or the front raises the risk of sudden infant death. About three thousand five hundred infants in the United States die each year from sleep-related causes, according to the 2022 AAP statement. The AAP advises that infants with reflux sleep on their backs on a firm, flat, non-inclined surface, and states that elevating the head of the crib is not effective in reducing reflux and is not recommended. Inclined sleep products have been deadly. The Consumer Product Safety Commission recorded ninety-seven deaths linked to inclined sleepers, one popular rocking sleeper was recalled in 2019 after about four point seven million had been sold, and the Safe Sleep for Babies Act of 2022 banned inclined sleepers with an angle above ten degrees. The reassurance families need is that healthy infants placed on their backs are not more likely to choke on spit-up, because their airway reflexes protect them. Both the joint guideline and NICE advise against positional treatment of reflux in sleeping infants. When the infant is awake and observed, holding him upright after a feed and supervised tummy time are safe and reasonable.

Part seven. What does not work, and the breastfed infant.

Several popular measures lack evidence. Infant massage was examined in one small trial in the 2018 guideline review, without benefit for reflux, and is not recommended for that purpose. Infant alginate was studied by Del Buono and colleagues in 2005 in twenty bottle-fed infants, using impedance and pH monitoring with alternating doses of alginate and placebo. Reflux episodes were about one point six per hour with either, acid reflux was not significantly different, and only the height of reflux fell slightly, from about seventy-seven to sixty-seven percent of the oesophageal length. NICE allows a one- to two-week alginate trial when its other steps have failed, while the joint guideline advises against chronic use. Infant seats and car seats after feeds increase reflux. Wedges, inclined sleepers and raised cots do not reduce reflux and are dangerous. Starting solid food before four months has no evidence for reflux and is linked to later obesity. Changing formula brands again and again has never been tested and delays the real diagnosis; one planned milk trial is better.

For the breastfed infant, NICE begins with a breastfeeding assessment by someone trained to do it: latch, positioning, frequency, oversupply and a fast let-down. Thickening is not a first step, because breast milk amylase digests starch, and carob can be used only with expressed milk. If allergy is suspected, the mother removes all cow's milk protein for two to four weeks, with calcium and vitamin D, followed by re-challenge. Only then, if needed, NICE allows an alginate trial, and any acid suppression belongs to a specialist. The trap to avoid is stopping breastfeeding to see whether formula helps, which turns a diagnostic question into a permanent change and removes the protection of breast milk.

Part eight. The stepped plan and the guidelines.

Put together, the plan is a series of single steps, each with a review date. First, look for red flags with a history, examination and growth chart; any warning sign takes the infant off this pathway. Second, explain the natural history. Third, correct excess volume and try smaller, more frequent feeds, reviewing in one to two weeks. Fourth, for formula-fed infants, use an anti-regurgitation formula or an appropriate thickener for about two weeks. Fifth, run a two- to four-week milk trial with a planned re-challenge. Only sixth, and only through a specialist, consider a time-limited trial of acid suppression. Change one thing at a time; changing feed volume, formula and position on the same day makes every later decision impossible.

How do the documents compare? On avoiding overfeeding, smaller and more frequent feeds, and thickened feeds, NASPGHAN and ESPGHAN and NICE are aligned, although neither gives a number for overfeeding, which is a gap. On sleep position they are also aligned with the AAP: supine and flat. They differ on the milk trial, because NICE has no hydrolysate step within its reflux pathway and reaches allergy through a separate route, and they differ on alginate, which NICE allows as a short trial and the joint guideline advises against for chronic use. And there is a material gap on thickeners in preterm infants: the xanthan warning lives in a regulatory alert rather than in the general reflux guidance that a family doctor of a neonatal

unit graduate is likely to read.

Several traps recur. Cutting milk to cut spilling can cause faltering growth. Believing that a thickened feed fixes reflux ignores that crying is unchanged and that cereal adds calories. A dairy-free diet without a re-challenge becomes a lifelong label. Raising the cot is ineffective and inclined products kill. And a xanthan thickener in a former preterm infant can cause NEC. Two more: changing several things at once, and treating a red flag such as green vomit, projectile vomiting or onset after six months with feeding advice.

What to do differently in clinic.

Consider a few infants. A three-month-old taking two hundred millilitres per kilogram per day, spilling after most feeds and gaining above the seventy-fifth centile, needs an appropriate daily total split into smaller feeds, with a weight check in two weeks. A formula-fed two-month-old with troublesome spilling, normal volumes and good growth can have a pre-thickened formula for two weeks, with the explanation that it treats the spills and not the crying. A thriving breastfed six-week-old with a fast let-down needs a breastfeeding assessment and reassurance, not a thickener. A four-month-old with eczema, mucus in the stools and persistent spilling despite thickened formula should have two to four weeks of extensively hydrolysed formula, then a planned re-challenge. Parents who want a wedge or a raised cot should hear clearly: supine, flat, no positioners, and upright holding after feeds while awake. A former thirty-week infant at two months corrected whose family bought a xanthan thickener online should stop it at once. And an infant who has had smaller feeds, thickening and a milk trial and is still crying, now losing weight, has left the non-drug pathway and needs a fresh look for red flags, a growth review and a specialist assessment.

So tomorrow in clinic, measure the volume before you judge it, and let the growth chart decide. Use thickening for spilling, choose the agent by age and milk, prefer pre-thickened formula, and never use xanthan in a preterm infant. Run any milk trial as a test, with a start date, an end date and a re-challenge, and keep the mother breastfeeding. Keep every infant on their back on a flat surface for sleep, and say out loud that raising the cot does not help. Change one thing at a time, review every step, and keep the proton pump inhibitor in the cupboard.